

HW 6.8.1.b: Graphs of Polar Functions

Match each equation with one of the graphs shown.

1. $r = 2 + 2\cos(\theta)$

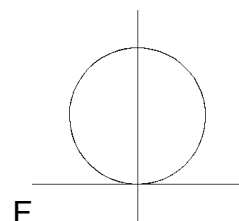
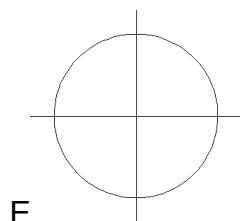
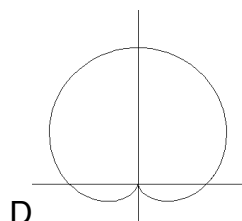
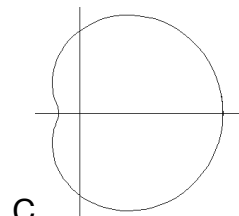
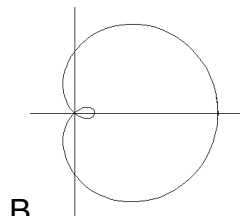
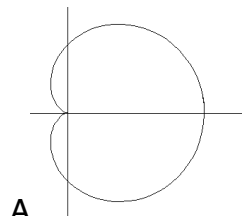
2. $r = 2 + 2\sin(\theta)$

3. $r = 4 + 3\cos(\theta)$

4. $r = 3 + 4\cos(\theta)$

5. $r = 5$

6. $r = 2\sin(\theta)$



Sketch a graph of the polar equation.

7. $r = 3\cos(\theta)$

8. $r = 2\sin(\theta)$

9. $r = 3\sin(2\theta)$

10. $r = 4\cos(3\theta)$

11. $r = 5\sin(3\theta)$

12. $r = 5\cos(4\theta)$

13. $r = 3\cos(2\theta)$

14. $r = \cos(5\theta)$

15. $r = 2 + 2\cos(\theta)$

16. $r = 3 + 2\sin(\theta)$

17. $r = 1 + 3\sin(\theta)$

18. $r = 2 + 4\cos(\theta)$

19. $\theta = \frac{\pi}{3}$

20. $\theta = \frac{7\pi}{6}$

21. $r = 3\sec\theta$

22. $r = -3\csc\theta$

23. $r = 4 + 2\cos\theta$

24. $r = 7$

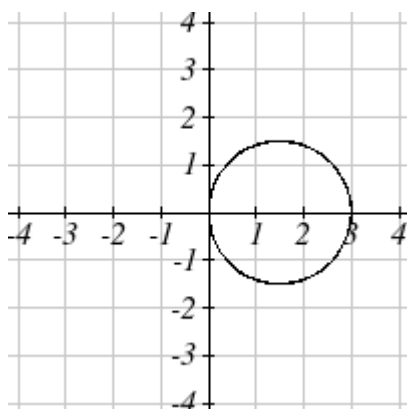
Selected Answers:

1. A

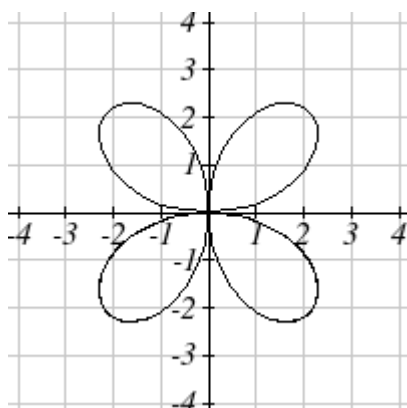
3. C

5. E

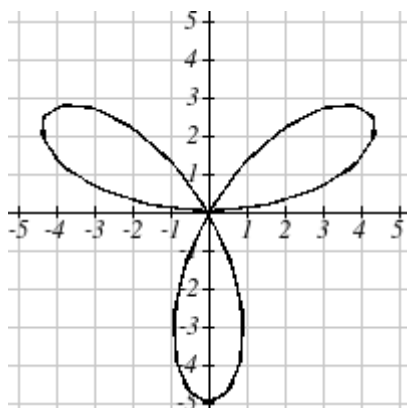
7.



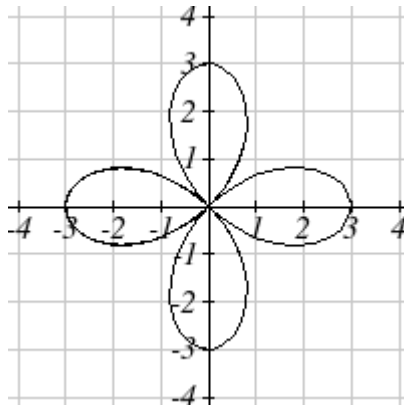
9.



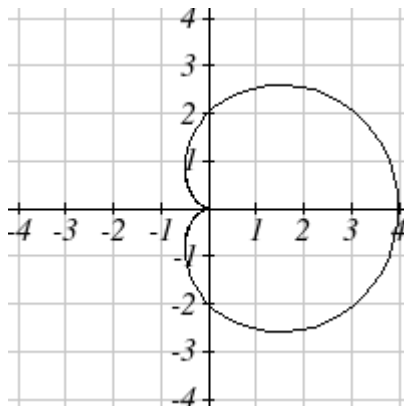
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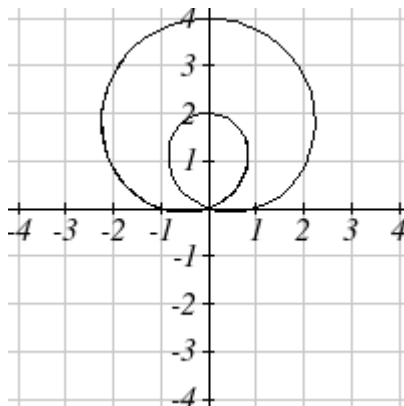
13.



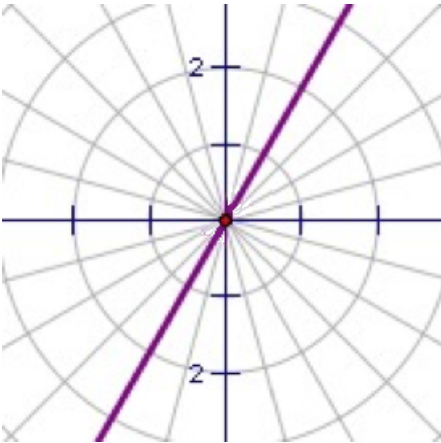
15.



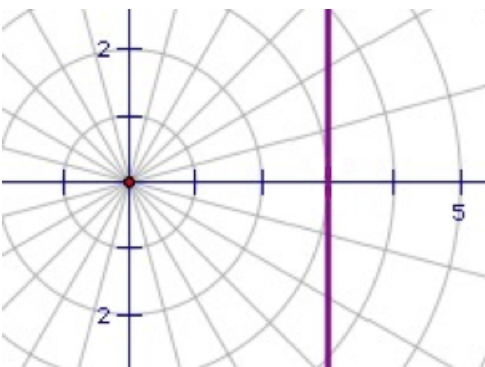
17.



19.



21.



23.

